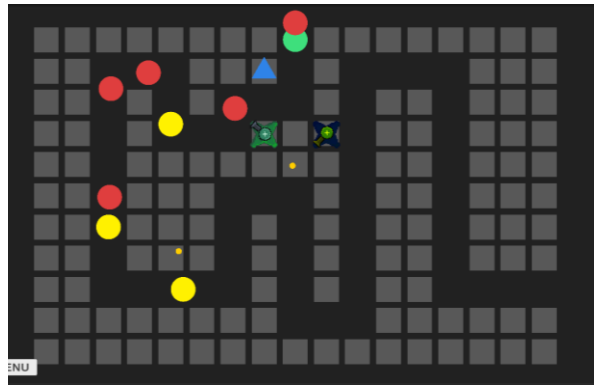


Tower Defense Game



Osman Furkan Sezgin 190709026

Muğla Sıtkı Koçman University

Computer Engineering

CENG 4011 Senior Design Project

20/01/2024

Tower Defense Game

Osman Furkan Sezgin
Computer Engineering
Muğla Sıtkı Koçman University
20/01/2024

Summary

In this project I made a tower defense game using Unity Game Engine. In this game there are different types of turrets that player can purchase from shop with earned money from killing enemies. There are three types of enemy and three types of turret. Each enemy and turrets has their own characteristics.

Players' goal is the killing all enemies before they reach base. If players fail they have to try again to reach next level.

Table of Contents

1. Introduction	2
2. Methods	2
2.1 Game Engine	Error! Bookmark not defined.
2.2 Art	Error! Bookmark not defined.
2.3 Level Design.....	3
2.4 Turrets	3
2.5 Enemies	4
2.6 User Interface	5
3. Results	5
4. Conclusions	5
5. References	6

1. Introduction

Tower defense (TD) is a subgenre of strategy games where the goal is to defend a player's territories or possessions by obstructing the enemy attackers or by stopping enemies from reaching the exits, usually achieved by placing defensive structures on or along their path of attack.[1] This typically means building a variety of different structures that serve to automatically block, impede, attack or destroy enemies.

In this project I made a tower defense game that is easy enough for casual players and challenging for more competitive players. People can enjoy the game by playing different levels or challenge themselves in challenge levels and compete other players for a spot on leaderboards.

2. Methods

I used Unity Game Engine to make the game. Turrets are made in pixel art style using Aseprite.

2.1 Game Engine

Unity is a cross-platform game engine developed by Unity Technologies, first announced and released in June 2005 at Apple Worldwide Developers Conference as a Mac OS X game engine. The engine has since been gradually extended to support a variety of desktop, mobile, console and virtual reality platforms. It is particularly popular for iOS and Android mobile game development, is considered easy to use for beginner developers, and is popular for indie game development.

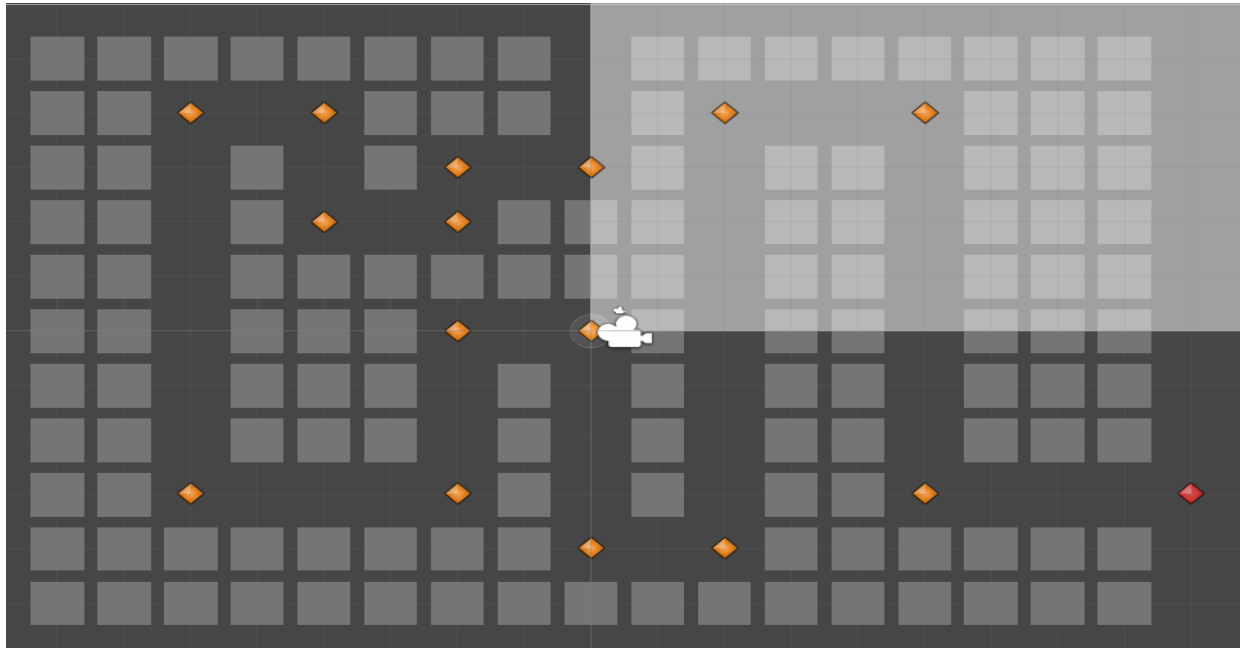
The engine can be used to create three-dimensional (3D) and two-dimensional (2D) games, as well as interactive simulations and other experiences.

2.2 Art

Aseprite is a proprietary, source-available image editor designed primarily for pixel art drawing and animation. It runs on Windows, macOS, and Linux, and features different tools for image and animation editing such as layers, frames, tilemap support, command-line interface, Lua scripting, among others. It is developed by Igará Studio S.A. and led by the developers David, Gaspar, and Martín Capello. have one sub-subheading.

2.3 Level Design

Each level has its own pattern and enemy waves. Levels become more difficult as the player advances to new levels.



There are two types of levels. First type is normal levels. Players unlock new levels by beating previous levels. Second type is challenge levels. Players unlock these levels after they unlock all normal levels. Challenge levels never ends until players fail and each level have their own leaderboard. Players try to collect most points and get on the leaderboard.

2.4 Turrets

There are three types of turrets in this game.

First one is basic turret (Figure 1). Basic turrets deal average damage at average attack speed. Players start levels with one basic turret money and place it wherever they want from placement spots.

Second one is sniper turret (Figure 2). Sniper turret is more expensive than basic turrets ,but it deals more damage than basic turret at slower attack speed. Sniper turret is strong against enemies with higher health but weak against enemies with fast movement speed. Player should decide whether if they need sniper turret before they buy them.

Third one is slow motion turret (Figure 3). Slow motion turret deals no damage, but it slows enemies to help other turrets to kill enemies easily. Slow motion turret is more expensive than basic turret and cheaper than sniper turret.



Figure 1



Figure 2

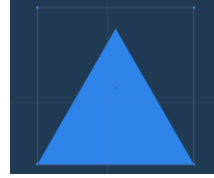


Figure 3

Turrets can be upgraded. Turret upgrades are not permanent and for a single session only. Each time an upgrade is purchased, the upgrade price for that turret increases. Upgrades increase turrets attack speed and damage for basic and sniper turret. Slow motion turret upgrades increase slow rate.

2.5 Enemies

There are three types of enemies in this game.

First one is basic enemy (Figure 4). Basic enemies move average speed, and they have average health. Basic turrets drop average amount of gold when they killed.

Second one is tank enemy (Figure 5). Tank enemies move slower than basic enemies, but they have more health. They are weak against sniper turrets but strong against basic turrets. Tank enemies drop more than average amount of gold when they killed.

Third one is fast enemy (Figure 6). Fast enemies move faster than basic enemies, but they have less health. Fast enemies strong against sniper turrets but weak against basic turrets. Fast enemies drop less than average amount of gold when they killed.

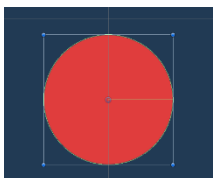


Figure 4

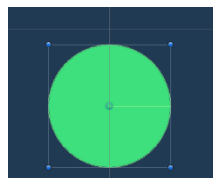


Figure 5

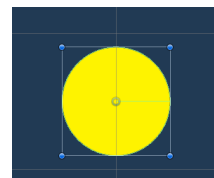


Figure 6

Enemies are getting stronger each wave. Their health increases each level and they become harder to kill.

2.6 User Interface

There is a menu (Figure 7) on the left side of the screen that can be opened and closed. Players can buy the tower they want from this menu if they can afford it.

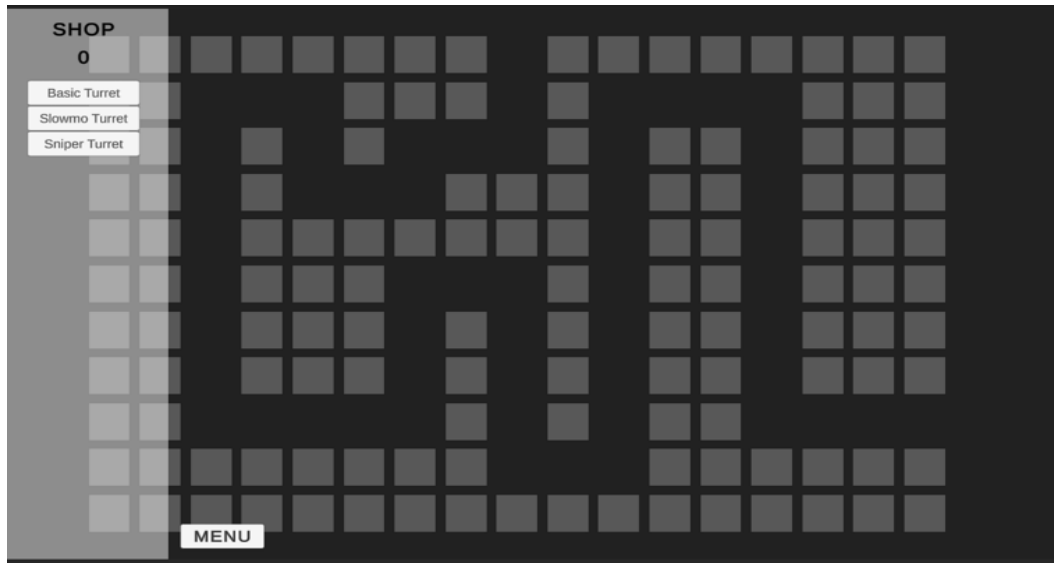


Figure 7

3. Results

I made a game that players can play and beat levels and challenge themselves in challenge levels.

4. Conclusions

On this project I made a game that is free to play so to make money from this game I will add Google Ads to game. Google Ads work better if you have more players. I am planning to add new levels, turrets and enemies so can players spend more time in this game and see more ads. I will try to use the most player friendly ad placements to avoid disturbing players.

I will renew the visuals and I will add music to game, which is currently the biggest deficiency in the game to make the game more playable.

Once I've done all this, I'll be ready to release the game.

5. References

How to make a Tower Defense Game YouTube tutorial by Brackeys.

2D Tower Defense Tutorial Series YouTube tutorial by Muddy Wolf.